# Damage-Response Network Observatory

A staged, nonclinical study of extracellular signaling, secondary senescence, glial feedback, and measurable restoration | Version 1.0

**STUDY CLASSIFICATION** Computational and observational first. No animal or patient experiments are proposed here. Any follow-on wet-lab manipulation requires qualified institutional oversight and a separate approved SOP.

## Study objective and primary question

Determine whether a cell-type-specific extracellular signaling graph explains secondary senescence-associated and tissue-function outcomes better than cell-intrinsic molecular measurements alone, and whether the graph predicts resolution versus persistence. The first test uses public data from the published human brain-cell model, not fabricated evidence of direct chemical exchange. [R1]

## 1 / Preregistered hypotheses

| Code | Hypothesis and falsification criterion |
|---|---|
| H1 — Sender specificity | Astrocyte/microglia sender identity plus measured state predicts recipient changes. Fails if cell-type labels alone perform equally well on held-out biological replicates. |
| H2 — Ligand/receptor value | A vetted extracellular ligand–receptor feature set improves predictions of recipient endpoints over intrinsic-only models. Fails if improvement disappears in held-out groups or shuffled-edge controls. |
| H3 — Persistence | A feedback-aware model distinguishes resolving from persistent responses where repeated measurements exist. Mark untestable if the data lack time series. |
| H4 — Restoration signal | In independent validated data, a lower damage-related state corresponds to preserved or improved cell function without broad loss of protective activity. Fails if improved marker values coincide with reduced viability or function. |

## 2 / Study design: five gated phases

| Phase | Workstream | Required output / decision |
|---|---|---|
| G0 | Data audit and evidence registry | Confirm E-MTAB-15005 files, design, cell types, labels and replicate units; catalogue published protein results separately. |
| G1 | Cell-state profiles | Estimate relative SASP/senescence-associated patterns by condition, cell type and donor/sample. |
| G2 | Directed candidate map | Build sender–ligand–receptor–recipient graph; distinguish measured, inferred and hypothesized edges. |
| G3 | Out-of-sample tests | Compare intrinsic-only vs extracellular-augmented prediction, with sample-level cross-validation and null controls. |
| G4 | Restoration simulation | Model signal attenuation and recovery as counterfactual hypotheses; no claims of actual therapy. |

## 3 / Starting data, measurements, and traceability

Primary source: Russo & Riessland (2026), RNA-seq ArrayExpress E-MTAB-15005. Published experiments tested five human brain-related cell lines: astrocytes, microglia, neurons, oligodendrocytes, and endothelial cells. The paper reports extracellular SASP-associated proteomic/cytokine comparisons and recipient state effects; some additional datasets may require author permission. [R1]

| Data class | Minimum columns or assays | Evidence role |
|---|---|---|
| Sample metadata | sample_id, biological_replicate_id, cell_type, donor_state, recipient_state, time, batch, assay | Independent units / confound control |
| Cell-intrinsic RNA | gene_id, count, normalization, QC; CDKN1A, CDKN2A, LMNB1 and broad state signatures | Cell-state association |
| Extracellular signaling | molecule_id, source-cell type, concentration or measured secretion result, assay, replicate | Sender evidence when actually measured |
| Receptors | recipient-cell receptor RNA/protein, specific ligand pair, evidence source | Predicted interface unless experimentally validated |
| Functional endpoints | SA-β-gal, viability, mitochondrial indicators, synaptic support when available | Damage versus recovery |

## 4 / Evidence graph definition

Each graph edge must carry provenance and an evidence tier. For example: astrocyte → CCL2-associated signal → receptor-bearing receiver → recipient molecular state. Do not assert that any particular receptor was activated unless measured. Keep ATP–P2X7 and microglia IL-1α/TNF/C1q pathways as independently sourced candidate layers, not as if all were experimentally tested in E-MTAB-15005. [R1, R2, R6]

| Tier | Minimum requirement | Allowed interpretation |
|---|---|---|
| E0 / Hypothesis | Literature suggests a potential ligand/receptor relationship | Candidate only |
| E1 / Co-expression | Ligand RNA in sender and receptor RNA in receiver | Plausible interface; no proof of signaling |
| E2 / Chemical support | Measured secreted factor and context-compatible recipient response | Supported association |
| E3 / Directional effect | Published donor → receiver exposure with matched controls | Experimental response in that model |
| E4 / Causal confirmation | Independent intervention changes receiver outcome without unacceptable off-target harm | Mechanistic support, not clinical efficacy |

## 5 / Computational analysis and controls

1. Audit accession contents and publisher supplements; record licenses, tissue context, original inclusion rules, independent replicate identity, assay coverage, and missing data.
2. Normalize and quality-check expression with documented thresholds, summarize at biological-replicate level, and preserve original raw counts.
3. Create cell-state signatures without calling any single p16/p21/SA-β-gal endpoint definitive senescence; assess tissue context and co-occurring viability markers.
4. Integrate published secretome results only where source and assay are documented; never impute actual measured secretion from RNA alone.

5. Infer candidate ligand–receptor links with curated references (e.g., CellChat), label them predicted, and form a cell-type directed multigraph. [R5]
6. Evaluate baseline A: cell type + batch + intrinsic RNA; B: A + candidate extracellular features; C: B + time/spatial features if truly available.
7. Hold out entire biological replicates or preparations; use group-aware cross-validation, permutation/null-edge tests, confidence intervals and multiplicity control.
8. Perform sensitivity tests removing each nominated pathway and assess whether effects survive independent batch/cohort evaluation.

## 6 / Outcomes and decision rules

Primary endpoint: held-out prediction improvement for defined recipient-cell state or functional measures after adding vetted extracellular features. Report effect size, confidence interval, calibration and negative controls. A prespecified minimum useful improvement must be chosen from pilot variability before the confirmatory test; do not invent a biological threshold.

| Outcome | Interpretation criteria |
|---|---|
| Positive computational result | Extracellular feature set adds reproducible out-of-sample value and survives replicate-level and shuffled-network controls. |
| Negative result | No reliable value beyond intrinsic cell state, or results disappear after confound adjustment. |
| Ambiguous | Insufficient replicates, assay mismatch, missing measured extracellular signals, non-independent cells, or poor calibration. |
| Proposed restoration | Lower propagation-associated signature PLUS preserved or improved function and viability in independently validated models. |

## 7 / Alternative explanations

- The recipient change could reflect nonspecific stress, culture medium differences, altered nutrient content, contamination, or cell loss rather than a transmitted damage program.
- A ligand may be elevated in a donor yet not activate a receptor in recipients; receptor RNA may not predict membrane protein abundance.
- An inflammatory response could be protective and self-limiting; removing it might worsen outcomes.
- An algorithm could learn batch, treatment, or cell-line identity instead of transferable mechanisms; model leakage is especially likely if single cells are treated as independent replicates.
- Published associations may not generalize to intact mouse or human brain tissue; no behavioral or clinical inference is permitted from cell culture alone.

## 8 / Restricted follow-on validation (separate approval)

If the computational study yields reproducible candidate edges, an authorized institutional group may design a distinct noninfectious, nonclinical validation using precharacterized neuron–astrocyte–microglia systems and matched controls. Compare baseline cell states, documented donor/recipient communication, and validated functional endpoints. The current protocol intentionally does not specify cell-injury induction conditions, chemical doses, genetic constructs, invasive procedures, or animal dosing. Independent safety review is required before any biological work.

## 9 / ChristEye Living Periodic Table mapping

| Category | Operational variable | Restoration test |
|---|---|---|
| Marrow | Inherited molecular programs and measured | Stable identity correctly controlled |

| Category | Operational variable | Restoration test |
|---|---|---|
| | transcriptional background | |
| Lotus | Cellular signal integration and response regulation | Return of context-appropriate responses |
| Winds | Extracellular secretion, diffusion and clearance | Measured harmful-signal persistence declines |
| Joints | Documented ligand–receptor pathways | Pathway-specific rather than nonspecific effect |
| Roots | Persistent cellular programs and repeated responses | Response returns toward healthy baseline over time |
| Heart | Cell survival, homeostatic and tissue-level outcomes | Improved function and maintained defense |
| Understanding | Reliable adaptive response, explicitly measured | Better held-out function; no spiritual claim |

**INTERPRETATION BOUNDARY** "Spirit of Error" and "repair of souls" remain philosophical labels within ChristEye. This study measures biochemical communication, sustained dysfunction, and functional recovery; it cannot establish spiritual causation, conscious cellular intentions, or a universal decay network.

## 10 / Deliverables and stopping criteria

- Accession inventory, data dictionary, literature evidence register, analysis notebook, reproducible versioned graph, null-control report and negative-result log.
- A ranked list of candidate sender–receiver edges annotated by evidence tier and uncertainty.
- A reproducible comparison showing whether extracellular features add predictive power, with independent biological units.
- Stop short of therapeutic claims unless a separate causal and safety program demonstrates functional benefit.

## Scientific references and data

**[R1]** Russo T, Riessland M. Characterizing the SASP-Dependent Paracrine Spreading of Senescence Between Human Brain Cell Types. Aging Cell. 2026;25:e70673. doi:10.1111/acel.70673. RNA-seq accession: E-MTAB-15005.

**[R2]** Liddelow SA et al. Neurotoxic reactive astrocytes are induced by activated microglia. Nature. 2017;541:481-487. doi:10.1038/nature21029.

**[R3]** Escartin C et al. Reactive astrocyte nomenclature, definitions, and future directions. Nature Neuroscience. 2021;24:312-325. doi:10.1038/s41593-020-00783-4.

**[R4]** Xu M et al. Senolytics improve physical function and increase lifespan in old age. Nature Medicine. 2018;24:1246-1256. doi:10.1038/s41591-018-0092-9.

**[R5]** Jin S et al. CellChat for systematic analysis of cell-cell communication from single-cell transcriptomics. Nature Protocols. 2024. doi:10.1038/s41596-024-01045-4.

**[R6]** P2X7 Signaling in Neuroinflammation: Glial Crosstalk, Redox Integration, and Therapeutic Targeting. 2026 review, PubMed PMID:42573850. Review-level synthesis, not proof of one universal pathway.